Bihar Engineering University, Patna
End Semester Examination - 2023

Course: B.Tech. | *Semester-VI* | *Time: 03 Hours*
Code: 103605 | *Subject: High Voltage Engineering* | *Full Marks: 70*

Instructions:-

(i) The marks are indicated in the right-hand margin.
(ii) There are NINE questions in this paper.
(iii) Attempt FIVE questions in all.
(iv) Question No. 1 is compulsory.

Q.1 ***Write answer of the following (Any seven question only):*** **[2 x 7 = 14]**

(a) Define Townsend's theory.
(b) Define impulse voltage.
(c) Why transmission occurs on high voltage?
(d) List the standards for HV testing of electrical apparatus.
(e) Define switching overvoltage.
(f) Mention the role of dielectric in high voltage engineering.
(g) Name various tests to be carried out on circuit breaker.
(h) Define Surge impedance loading.
(i) Define lightning phenomenon
(j) List out various schemes of protection against over voltages.

Q.2 (a) Give the origin and characteristics of switching surges and explain the causes of over voltage due to switching surges in EHV and UHV system. **[7]**
(b) Describe the mechanism of breakdown in composite insulation in detail. **[7]**

Q.3 (a) Explain the different methods of producing switching impulses in test laboratories **[7]**
(b) Discuss elaborately the principle and operation of impulse current generator **[7]**

Q.4 What are the tests to be conducted on cables as IS 10810. Explain them in detail. Discuss the different high voltage tests conducted on bushings **[14]**

Q.5 What are the tests conducted on isolators and circuit breakers? Explain in detail. **[14]**

Q.6 Define Townsend's first and second ionization mechanism co-efficient. Derive an expression for the current in the air gap considering Townsend secondary ionization coefficient. **[14]**

Q.7 (a) Explain one method of controlled tripping of impulse generators. Why is controlled tripping necessary? **[7]**
(b) Three 350 KV, 350 KVA testing transformers are connected in cascaded and have a short-circuit impedance of 5%. Determine (i) the full-load current (ii) the short-circuit current (iii) the maximum capacitive load that can be tested without exceeding the power rating. **[7]**

Q.8 (a) What is stressed oil volume theory? How does it explain breakdown in large volumes of commercial liquid dielectrics? **[7]**
(b) What is thermal breakdown in solid dielectrics and how is it practically more significant than other mechanisms? **[7]**

Q.9 Write short notes on *any two* of the following: **[7 x 2 =14]**

(a) Surge diverters
(b) Application of insulating materials
(b) Generation of high d.c. voltage
(d) Moisture content in the oil

© Apna BEU
https://www.akubihar.com
Apna BEU: Syllabus, PYQs & AI
All-in-One app for Syllabus, PYQs, Result, Attendance and more.
GET IT ON Google Play